2VG3 Homework 4 - RigidBody Collisions in 2D

Generated from rigidbody2d_collision.py using the assignment scenarios 0-4 with e=1 and e=0. Collision points are averaged when multiple simultaneous contacts are present.

Exercise 2 (a) - Collision Details with e=1.0

Scenario 0: Default setup, no initial rotation.

```
Collision 0: time=2.504166666666686, point=[-0.004171311855316162, 25.0, 0.0], normal=[1.0, 0.0, -0.0],
mode=simultaneous-contacts-averaged, contacts_before_average=4
```

Simultaneous contacts detected and averaged to one collision point. contact 0: point (0.008328557014465332, 25.0, 0.5), type body 0 corner against body 1 face | contact 1: point (-0.016671180725097656, 25.0, -0.5), type body 1 corner against body 0 face | contact 2: point (-0.016671180725097656, 25.0, 0.5), type face-face (edge-edge) contact | contact 3: point (0.008328557014465332, 25.0, -0.5), type face-face (edge-edge) contact

Scenario 1: Body 0 starts at 30 degrees.

```
Collision 0: time=2.441666666666683, point=[0.23867040872573853, 25.0, -0.14014281332492828],
normal=[1.0, 0.0, -0.0], mode=simultaneous-contacts-averaged, contacts_before_average=3
```

Simultaneous contacts detected and averaged to one collision point. contact 0: point (0.24935407936573029, 25.0, -0.1339745968580246), type body 0 corner against body 1 face | contact 1: point (0.23332858085632324, 25.0, -0.12472227215766907), type face-face (edge-edge) contact | contact 2: point (0.23332858085632324, 25.0, -0.1617315709590912), type face-face (edge-edge) contact

Scenario 2: Body 0 starts at 60 degrees, body 1 at 30 degrees.

```
Collision 0: time=2.475000000000018, point=[-0.2628963887691498, 25.0, 0.1339745968580246],
normal=[0.8660253882408142, 0.0, 0.5], mode=simultaneous-contacts-averaged, contacts_before_average=3
```

Simultaneous contacts detected and averaged to one collision point. contact 0: point (-0.26603004336357117, 25.0, 0.1339745968580246), type body 1 corner against body 0 face | contact 1: point (-0.2636798024177551, 25.0, 0.1380452960729599), type face-face (edge-edge) contact | contact 2: point (-0.25897935032844543, 25.0, 0.1299038976430893), type face-face (edge-edge) contact

Scenario 3: Body 0 starts with omega=-5 rad/s; body 1 starts at x=-1.9,z=1.9 with zero velocity.

```
Collision 0: time=1.6999999999999953, point=[-2.4036319255828857, 25.0, 0.90016633272171O2],
normal=[-0.0, 0.0, 1.0], mode=simultaneous-contacts-averaged, contacts_before_average=3
```

Simultaneous contacts detected and averaged to one collision point. contact 0: point (-2.403536558151245, 25.0, 0.9004989862442017), type body 0 corner against body 1 face | contact 1: point (-2.404198408126831, 25.0, 0.8999999761581421), type face-face (edge-edge) contact | contact 2: point (-2.403160333633423, 25.0, 0.8999999761581421), type face-face (edge-edge) contact

Scenario 4: Body 0 starts with omega=+3 rad/s; body 1 starts at -10 degrees.

```
Collision 0: time=2.516666666666866, point=[0.12117062509059906, 25.0, -0.6363198161125183],
normal=[0.9541522860527039, 0.0, 0.29932165145874023], mode=simultaneous-contacts-averaged,
contacts_before_average=3
```

Simultaneous contacts detected and averaged to one collision point. contact 0: point (0.12216929346323013, 25.0, -0.6584559082984924), type body 1 corner against body 0 face | contact 1: point (0.11072244495153427, 25.0, -0.5935376286506653), type face-face (edge-edge) contact | contact 2: point (0.13062013685703278, 25.0, -0.6569657921791077), type face-face (edge-edge) contact

```
Collision 1: time=2.5458333333333547, point=[-0.16001787781715393, 25.0, 0.5922383666038513],
normal=[0.9638697504997253, 0.0, 0.26637396216392517], mode=simultaneous-contacts-averaged,
contacts_before_average=3
```

Simultaneous contacts detected and averaged to one collision point. contact 0: point (-0.16210471093654633, 25.0, 0.6268060803413391), type body 0 corner against body 1 face | contact 1: point (-0.17281149327754974, 25.0, 0.6250236630439758), type face-face (edge-edge) contact | contact 2: point (-0.14513741433620453, 25.0, 0.5248854160308838), type face-face (edge-edge) contact

Exercise 2 (b) - Collision Details with e=0.0

Scenario 0: Default setup, no initial rotation.

Per instructions for e=0, only the first contact properties are required.

```
Collision 0: time=2.504166666666686, point=[-0.004171311855316162, 25.0, 0.0], normal=[1.0, 0.0, -0.0],
mode=simultaneous-contacts-averaged, contacts_before_average=4
```

Simultaneous contacts detected and averaged to one collision point. contact 0: point (0.008328557014465332, 25.0, 0.5), type body 0 corner against body 1 face | contact 1: point (-0.016671180725097656, 25.0, -0.5), type body 1 corner against body 0 face | contact 2: point (-0.016671180725097656, 25.0, 0.5), type face-face (edge-edge) contact | contact 3: point (0.008328557014465332, 25.0, -0.5), type face-face (edge-edge) contact

Scenario 1: Body 0 starts at 30 degrees.

Per instructions for e=0, only the first contact properties are required.

```
Collision 0: time=2.441666666666683, point=[0.23867040872573853, 25.0, -0.14014281332492828],
normal=[1.0, 0.0, -0.0], mode=simultaneous-contacts-averaged, contacts_before_average=3
```

Simultaneous contacts detected and averaged to one collision point. contact 0: point (0.24935407936573029, 25.0, -0.1339745968580246), type body 0 corner against body 1 face | contact 1: point (0.23332858085632324, 25.0, -0.12472227215766907), type face-face (edge-edge) contact | contact 2: point (0.23332858085632324, 25.0, -0.1617315709590912), type face-face (edge-edge) contact

Scenario 2: Body 0 starts at 60 degrees, body 1 at 30 degrees.

Per instructions for e=0, only the first contact properties are required.

```
Collision 0: time=2.475000000000018, point=[-0.2628963887691498, 25.0, 0.1339745968580246],
normal=[0.8660253882408142, 0.0, 0.5], mode=simultaneous-contacts-averaged, contacts_before_average=3
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Scenario 3: Body 0 starts with omega=-5 rad/s; body 1 starts at x=-1.9,z=1.9 with zero velocity.

Per instructions for e=0, only the first contact properties are required.

```
Collision 0: time=1.6999999999999953, point=[-2.4036319255828857, 25.0, 0.90016633277217102],
normal=[-0.0, 0.0, 1.0], mode=simultaneous-contacts-averaged, contacts_before_average=3
```

Simultaneous contacts detected and averaged to one collision point. contact 0: point (-2.403536558151245, 25.0, 0.9004989862442017), type body 0 corner against body 1 face | contact 1: point (-2.404198408126831, 25.0, 0.8999999761581421), type face-face (edge-edge) contact | contact 2: point (-2.403160333633423, 25.0, 0.8999999761581421), type face-face (edge-edge) contact

Scenario 4: Body 0 starts with omega=+3 rad/s; body 1 starts at -10 degrees.

Per instructions for e=0, only the first contact properties are required.

```
Collision 0: time=2.5166666666666866, point=[0.12117062509059906, 25.0, -0.6363198161125183],
normal=[0.9541522860527039, 0.0, 0.29932165145874023], mode=simultaneous-contacts-averaged,
contacts_before_average=3
```

Simultaneous contacts detected and averaged to one collision point. contact 0: point (0.12216929346323013, 25.0, -0.6584559082984924), type body 1 corner against body 0 face | contact 1: point (0.11072244495153427, 25.0,

-0.5935376286506653), type face-face (edge-edge) contact | contact 2: point (0.13062013685703278, 25.0, -0.6569657921791077), type face-face (edge-edge) contact

Exercise 3 - Momentum, Angular Momentum, and Energy

Values are reported immediately before and after the first collision in each scenario.

Restitution e=1.0

Scenario	Stage	P	L	K_trans	K_rot	K_total
0	pre	(-2.0, 0.0, 0.0)	(0.0, -3.0, 50.0)	10.0	0.0	10.0
0	post	(-2.0, 0.0, 0.0)	(0.0, -3.0, 50.0)	2.859504595398903	7.140496293375321	10.000000888774224
1	pre	(-2.0, 0.0, 0.0)	(0.0, -3.0, 50.0)	10.0	0.0	10.0
1	post	(-2.0, 0.0, 0.0)	(0.0, -3.0, 50.0)	2.6182504184544086	7.381751049526937	10.000001467981345
2	pre	(-2.0, 0.0, 0.0)	(0.0, -3.0, 50.0)	10.0	0.0	10.0
2	post	(-2.0, 0.0, 0.0)	(0.0, -3.0, 50.0)	7.4552390575408936	2.544761038946728	10.000000096487621
3	pre	(2.0, 0.0, 0.0)	(0.0, -4.3333330154418945, -50.0)	2.0	8.333333333333332	10.333333333333332
3	post	(2.0, 0.0, 0.0)	(0.0, -4.3333330154418945, -50.0)	2.6486558616161346	7.684677500962694	10.333333362578829
4	pre	(-2.0, 0.0, 0.0)	(0.0, -1.0, 50.0)	10.0	3.0	13.0
4	post	(-2.0, 0.0, 0.0)	(0.0, -1.000000238418579, 50.0)	1.8837742805480957	11.116227188542748	13.000001469090844

Scenario 0: Delta_P=(0.0, 0.0, 0.0), Delta_L=(0.0, 0.0, 0.0). Total kinetic energy is conserved (within numerical tolerance). Energy transfers from bulk translation into rotation. Delta_K_total=8.887742239949148e-07. Delta_K_trans=-7.140495404601097. Delta_K_rot=7.140496293375321.

Scenario 1: Delta_P=(0.0, 0.0, 0.0), Delta_L=(0.0, 0.0, 0.0). Total kinetic energy is conserved (within numerical tolerance). Energy transfers from bulk translation into rotation. Delta_K_total=1.4679813453710722e-06. Delta_K_trans=-7.381749581545591. Delta_K_rot=7.381751049526937.

Scenario 2: Delta_P=(0.0, 0.0, 0.0), Delta_L=(0.0, 0.0, 0.0). Total kinetic energy is conserved (within numerical tolerance). Energy transfers from bulk translation into rotation. Delta_K_total=9.648762144820466e-08. Delta_K_trans=-2.5447609424591064. Delta_K_rot=2.544761038946728.

Scenario 3: Delta_P=(0.0, 0.0, 0.0), Delta_L=(0.0, 0.0, 0.0). Total kinetic energy is conserved (within numerical tolerance). Energy transfers from rotation into bulk translation. Delta_K_total=2.9245496691032713e-08. Delta_K_trans=0.6486558616161346. Delta_K_rot=-0.648655832370638.

Scenario 4: Delta_P=(0.0, 0.0, 0.0), Delta_L=(0.0, -2.384185791015625e-07, 0.0). Total kinetic energy is conserved (within numerical tolerance). Energy transfers from bulk translation into rotation. Delta_K_total=1.4690908436421068e-06. Delta_K_trans=-8.116225719451904. Delta_K_rot=8.116227188542748.

Restitution e=0.0

Scenario	Stage	P	L	K_trans	K_rot	K_total
0	pre	(-2.0, 0.0, 0.0)	(0.0, -3.0, 50.0)	10.0	0.0	10.0
0	post	(-2.0, 0.0, 0.0)	(0.0, -3.0, 50.0)	1.6694214083254337	1.7851240733438303	3.454545481669264
1	pre	(-2.0, 0.0, 0.0)	(0.0, -3.0, 50.0)	10.0	0.0	10.0
1	post	(-2.0, 0.0, 0.0)	(0.0, -3.0, 50.0)	1.7464055540040135	1.8454377623817342	3.5918433163857477
2	pre	(-2.0, 0.0, 0.0)	(0.0, -3.0, 50.0)	10.0	0.0	10.0
2	post	(-2.0, 0.0, 0.0)	(0.0, -3.0, 50.0)	3.324914038181305	0.636190259736682	3.961104297917987
3	pre	(2.0, 0.0, 0.0)	(0.0, -4.3333330154418945, -50.0)	2.0	8.333333333333332	10.333333333333332

Scenario	Stage	P	L	K_trans	K_rot	K_total
3	post	(2.0, 0.0, 0.0)	(0.0, -4.333333492279053, -50.0)	2.162163846194744	7.973465946674217	10.135629792868961
4	pre	(-2.0, 0.0, 0.0)	(0.0, -1.0, 50.0)	10.0	3.0	13.0
4	post	(-2.0, 0.0, 0.0)	(0.0, -1.0000001192092896, 50.0)	4.272379815578461	5.912237862456656	10.184617678035117

Scenario 0: Delta_P=(0.0, 0.0, 0.0), Delta_L=(0.0, 0.0, 0.0). Total kinetic energy decreases because the collision is inelastic (e=0). Energy transfers from bulk translation into rotation. Delta_K_total=-6.545454518330736. Delta_K_trans=-8.330578591674566. Delta_K_rot=1.7851240733438303.

Scenario 1: Delta_P=(0.0, 0.0, 0.0), Delta_L=(0.0, 0.0, 0.0). Total kinetic energy decreases because the collision is inelastic (e=0). Energy transfers from bulk translation into rotation. Delta_K_total=-6.408156683614252. Delta_K_trans=-8.253594445995986. Delta_K_rot=1.8454377623817342.

Scenario 2: Delta_P=(0.0, 0.0, 0.0), Delta_L=(0.0, 0.0, 0.0). Total kinetic energy decreases because the collision is inelastic (e=0). Energy transfers from bulk translation into rotation. Delta_K_total=-6.038895702082013. Delta_K_trans=-6.675085961818695. Delta_K_rot=0.636190259736682.

Scenario 3: Delta_P=(0.0, 0.0, 0.0), Delta_L=(0.0, -4.76837158203125e-07, 0.0). Total kinetic energy decreases because the collision is inelastic (e=0). Energy transfers from rotation into bulk translation. Delta_K_total=-0.19770354046437077. Delta_K_trans=0.1621638461947441. Delta_K_rot=-0.3598673866591149.

Scenario 4: Delta_P=(0.0, 0.0, 0.0), Delta_L=(0.0, -1.1920928955078125e-07, 0.0). Total kinetic energy decreases because the collision is inelastic (e=0). Energy transfers from bulk translation into rotation. Delta_K_total=-2.815382321964883. Delta_K_trans=-5.727620184421539. Delta_K_rot=2.9122378624566556.